
Problem Set 3

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PY 451 Section A1

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1 Representations and Basis States

1. Even though our space is infinite-dimensional, we can isolate v_1, v_2 to consider a 2-dimensional subspace. The most general possible choice in this subspace, as illustrated by equation (1.1), is

$$\begin{aligned}|1\rangle &= \cos \theta e^{i\phi_1} v_1 + \sin \theta e^{i\phi_2} v_2, \\ |2\rangle &= \sin \theta e^{i\phi_3} v_1 - \cos \theta e^{i(\phi_3+\phi_1-\phi_2)} v_2.\end{aligned}$$

2. Note that due to orthonormality, for $v \in \{|3\rangle, |4\rangle\}$,

$$\langle v | 1 \rangle = \frac{1}{2} \sum_i v_i = 0.$$

Similarly,

$$\langle v | 2 \rangle = \frac{1}{2}(v_1 + v_2 - v_3 - v_4) = 0.$$

Therefore, we have

$$v_1 + v_2 + v_3 + v_4 = v_1 + v_2 - v_3 - v_4 \implies v_3 = -v_4.$$

We then have $v_1 + v_2 = 0$ and thus $v_1 = -v_2$. We then have that $|3\rangle, |4\rangle$ must satisfy

$$\begin{pmatrix} v_1 \\ -v_1 \\ v_3 \\ -v_3 \end{pmatrix}.$$

These vectors are determined by two distinct components v_1, v_3 , and thus we can plug in equation 1.1 to arrive at

$$|3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} \cos \theta e^{i\phi_1} \\ -\cos \theta e^{i\phi_1} \\ \sin \theta e^{i\phi_2} \\ -\sin \theta e^{i\phi_2} \end{pmatrix}, \quad |4\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} \sin \theta e^{i\phi_3} \\ -\sin \theta e^{i\phi_3} \\ -\cos \theta e^{i(\phi_3-\phi_1+\phi_2)} \\ \cos \theta e^{i(\phi_3-\phi_1+\phi_2)} \end{pmatrix}.$$

The $1/\sqrt{2}$ is the normalizing factor, which I guess I will show the calculation for.

$$\begin{pmatrix} \cos \theta e^{-i\phi_1} & -\cos \theta e^{-i\phi_1} & \sin \theta e^{-i\phi_2} & -\sin \theta e^{-i\phi_2} \end{pmatrix} \begin{pmatrix} \cos \theta e^{i\phi_1} \\ -\cos \theta e^{i\phi_1} \\ \sin \theta e^{i\phi_2} \\ -\sin \theta e^{i\phi_2} \end{pmatrix} = \cos^2 \theta + \cos^2 \theta + \sin^2 \theta + \sin^2 \theta.$$

This is the norm squared, so the norm is $\sqrt{2}$. I'll calculate the next one as well:

$$\begin{pmatrix} \sin \theta e^{-i\phi_3} & -\sin \theta e^{-i\phi_3} & -\cos \theta e^{-i(\phi_3-\phi_1+\phi_2)} & \cos \theta e^{-i(\phi_3-\phi_1+\phi_2)} \end{pmatrix} \begin{pmatrix} \sin \theta e^{i\phi_3} \\ -\sin \theta e^{i\phi_3} \\ -\cos \theta e^{i(\phi_3-\phi_1+\phi_2)} \\ \cos \theta e^{i(\phi_3-\phi_1+\phi_2)} \end{pmatrix}$$

$$= \sin^2 \theta + \sin^2 \theta + \cos^2 \theta + \cos^2 \theta = 2.$$

I will also show that they are orthogonal to each other:

$$\begin{pmatrix} \cos \theta e^{-i\phi_1} & -\cos \theta e^{-i\phi_1} & \sin \theta e^{-i\phi_2} & -\sin \theta e^{-i\phi_2} \end{pmatrix} \begin{pmatrix} \sin \theta e^{i\phi_3} \\ -\sin \theta e^{i\phi_3} \\ -\cos \theta e^{i(\phi_3-\phi_1+\phi_2)} \\ \cos \theta e^{i(\phi_3-\phi_1+\phi_2)} \end{pmatrix}$$

$$= 2 \cos \theta \sin \theta e^{i(\phi_3-\phi_1)} + 2 \cos \theta \sin \theta e^{i(\phi_3-\phi_1)} = 2\langle + | - \rangle = 0.$$

These kets are also orthogonal to $|1\rangle, |2\rangle$ because they satisfy the property I listed earlier, and since this is a set of 4 linearly independent vectors in a 4 dimensional vector space, they are necessarily a basis.

2 Operators

1. Note that if A is Hermitian, then $\langle p|A|\Omega\rangle = \langle p|A^\dagger|\Omega\rangle$. By contrapositive, if $\langle p|A|\Omega\rangle \neq \langle p|A^\dagger|\Omega\rangle$, then A is not Hermitian. Note that

$$\langle p|A|\Omega\rangle = \langle p|0\rangle = 0.$$

We have

$$\langle p|A^\dagger|\Omega\rangle = (\langle \Omega|A|p\rangle)^\dagger = (\langle \Omega|0\rangle)^\dagger.$$

Since Ω is a state, it must satisfy $\langle \Omega|\Omega\rangle = 1$, and $1^\dagger = 1^* = 1$. Therefore, $\langle p|A|\Omega\rangle \neq \langle p|A^\dagger|\Omega\rangle$, and A is not Hermitian.

2. Note that $h(x)g(x) = 0$ (except possibly on $\{1/3, 2/3\}$ depending on how the drawing is interpreted, but this set has lebesgue measure 0 so the following conclusion still holds). It follows that

$$\int_0^1 h(y)g(y)g(x) \, dy = 0.$$

Therefore,

$$(Ah)(x) = \int_0^1 h(y)^2 h(x) \, dy + \int_0^1 h(y)v(y)v(x) \, dx.$$

Note that $v(x) = \frac{1}{3}h(x)$. It follows that

$$(Ah)(x) = \int_0^1 h(y)^2 h(x) \, dy + \int_0^1 \frac{1}{9} h(y)^2 h(x) \, dy = \frac{10}{9} h(x) \int_0^1 h(y)^2 \, dy.$$

Also note that since $0^2 = 0$ and $1^2 = 1$, we have that $h(x)^2 = h(x)$, so

$$(Ah)(x) = \frac{10}{9} h(x) \int_0^1 h(y) \, dy = \frac{20}{27} h(x).$$

Computing $(Ag)(x)$ is more simple, since the first and last integrals vanish. We have

$$(Ag)(x) = g(x) \int_0^1 g(y)^2 dy = g(x) \int_{1/3}^{2/3} dx = \frac{1}{3}g(x).$$

3. By part 2, $20/27$ is an eigenvalue with $h(x)$ as an eigenfunction, and $1/3$ is an eigenvalue with $g(x)$ as an eigenfunction, and any scalar multiple of these functions is also an eigenfunction. Note that the image of A must be a subset of the span of $\{h, g\}$ since $v = 1/3h$ is in this span, and thus these are clearly the only possible linearly independent eigenspaces other than the 0 eigenspace.

3 Born's Rule and Expectation Values

1. Recall:

$$S_{\vec{n}} = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix},$$

and the eigenvectors associated with eigenvalues $\hbar/2, -\hbar/2$ are

$$|\psi_1\rangle = \cos \frac{\theta}{2} |+\rangle + \sin \frac{\theta}{2} |-\rangle, \quad |\psi_2\rangle = \sin \frac{\theta}{2} |+\rangle - \cos \frac{\theta}{2} |-\rangle,$$

respectively. We have

$$\mathcal{P}_{\hbar/2} = |\langle + | \psi_1 \rangle|^2 = \left| \cos \frac{\theta}{2} \langle + | + \rangle + \sin \frac{\theta}{2} \langle + | - \rangle \right|^2 = \left| \cos \frac{\theta}{2} \right|^2.$$

Similarly, we have

$$\mathcal{P}_{-\hbar/2} = |\langle + | \psi_2 \rangle|^2 = \left| \sin \frac{\theta}{2} \langle + | + \rangle - \cos \frac{\theta}{2} \langle + | - \rangle \right|^2 = \left| \sin \frac{\theta}{2} \right|^2.$$

The former is the probability of measuring $\hbar/2$, since $|\psi_1\rangle$ is the corresponding eigenvector, and the latter is the probability of finding $-\hbar/2$ for the same reason. As a quick sanity check,

$$\left| \sin \frac{\theta}{2} \right|^2 + \left| \cos \frac{\theta}{2} \right|^2 = \sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} = 1.$$

2. (a)

$$\begin{aligned} \langle + | S_{\vec{n}} | + \rangle &= \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} \\ &= \frac{\hbar}{2} \cos \theta. \end{aligned}$$

- (b)

$$\langle S_{\vec{n}} \rangle = \sum_i \lambda_i \mathcal{P}_i = \frac{\hbar}{2} \cos^2 \frac{\theta}{2} - \frac{\hbar}{2} \sin^2 \frac{\theta}{2} = \frac{\hbar}{2} \left(\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \right) = \frac{\hbar}{2} \cos \theta.$$

3. (a) Note that

$$\mathcal{P}_{\hbar/2} = |\langle \psi_1 | \psi_1 \rangle|^2 \equiv 1.$$

It follows that $\mathcal{P}_{-\hbar/2} = 0$ by classical probability theory. Therefore, $|\psi_1\rangle$ and $|\psi_2\rangle$ are orthogonal states.

(b) i.

$$\begin{aligned} \langle \psi_1 | S_{\vec{n}} | \psi_1 \rangle &= \frac{\hbar}{2} \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} \cos \theta \cos \frac{\theta}{2} + \sin \theta \sin \frac{\theta}{2} \\ \cos \frac{\theta}{2} \sin \theta - \sin \frac{\theta}{2} \cos \theta \end{pmatrix} \\ &= \frac{\hbar}{2} \left(\cos \theta \cos^2 \frac{\theta}{2} + \sin \theta \cos \frac{\theta}{2} \sin \frac{\theta}{2} + \cos \frac{\theta}{2} \sin \frac{\theta}{2} \sin \theta - \sin^2 \frac{\theta}{2} \cos \theta \right) \\ &= \frac{\hbar}{2} \left(\cos \theta \left(\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \right) + 2 \sin \theta \cos \frac{\theta}{2} \sin \frac{\theta}{2} \right) \\ &= \frac{\hbar}{2} (\cos^2 \theta + \sin^2 \theta) \\ &= \frac{\hbar}{2}. \end{aligned}$$

ii.

$$\langle S_{\vec{n}} \rangle = \sum_i \lambda_i \mathcal{P}_i = \frac{\hbar}{2} 1 - \frac{\hbar}{2} 0 = \frac{\hbar}{2}.$$